# SLICING: AN LFP IMMERSION CUTTING THE SOURCE FIBRE DIMENSION BY ONE WHEN THE STABILIZER IS SMALLER

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*Allowed standard prerequisites (editorial).*

- Definition of a finite type point (morphisms chapter, Definition morphisms-definition-finite-type-point, canonical tag 02J1). cited at statement lines 2072-2073. The hypothesis on u is phrased with this notion.
- Notation situation for slicing: the restriction of the groupoid via g, the stabilizer group scheme G, the map h equal to s composed with the first projection, and the fibres F_u, G_u, F'_{u'} (more-groupoids chapter, Situation situation-slice, no canonical tag in the snapshot; no-tag pointer to the source label line). cited at statement lines 2075 and 2083. Fixes the notation d_1, d_2, h, F_u, G_u and the conclusion items in which it is phrased.
- A finite type point has an affine neighbourhood in which it is closed (morphisms chapter, Lemma morphisms-lemma-identify-finite-type-points, canonical tag 02J2). cited at proof lines 2095-2097. Provides the affine neighbourhood Spec(A) with u a closed point.
- Dimension of the fibre of a morphism at a point equals the dimension of the local ring of the fibre (morphisms chapter, Lemma morphisms-lemma-dimension-fibre-at-a-point, canonical tag 02FX). cited at proof lines 2128-2129 and 2229-2232. Identifies d_2 with the dimension of the local ring of the fibre F_u at e(u), and later evaluates the dimension of F'_u at e'(u) as the dimension of the cut fibre ring.
- Restriction of a groupoid preserves local finite type of the structure maps (more-groupoids chapter, Lemma lemma-restrict-preserves-type, no canonical tag in the snapshot; no-tag pointer to the source label line). cited at proof lines 2134-2137. The restriction of R to the point u is a groupoid scheme with locally finite type structure maps.
- For a groupoid on a field with locally finite type structure maps the dimensions of the restricted arrow scheme and the stabilizer agree (more-groupoids chapter, Lemma lemma-groupoid-on-field-dimension-equal-stabilizer, no canonical tag in the snapshot; no-tag pointer to the source label line).

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cited at proof lines 2138-2140. Gives dim(R") = dim(G") for the restriction to the point u.

- For such groupoids on a field the dimension equals the dimension of the local ring at the identity point (more-groupoids chapter, Lemma lemma-groupoid-on-field-locally-finite-type-dimension, no canonical tag in the snapshot; no-tag pointer to the source label line). cited at proof lines 2141-2142. Converts dim(R") into the dimension of the local ring of R" at e".
- The stabilizer of a restriction is the fibre of the stabilizer (groupoids chapter, Lemma groupoids-lemma-restrict-stabilizer, canonical tag 04ML). cited at proof lines 2143-2145. Identifies G" with G_u, so the dimension of the local ring of R" at e" equals d_1.
- Dimension formula for Cohen-Macaulay local rings and minimal primes (algebra chapter, Lemma algebra-lemma-CM-dim-formula, canonical tag 00NA). cited at proof lines 2181-2183. Each minimal prime of the Cohen-Macaulay fibre ring has quotient of dimension d_2.
- Prime avoidance (algebra chapter, Lemma algebra-lemma-silly, canonical tag 00DS). cited at proof lines 2192-2194. Produces f in the maximal ideal whose t-image avoids all the minimal primes.
- One equation cuts dimension (algebra chapter, Lemma algebra-lemma-one-equation, canonical tag 00KW). cited at proof lines 2195-2197. Gives the displayed inequality dim below d_2 for the fibre ring cut by f.
- Reformulation of Cohen-Macaulayness: over a Cohen-Macaulay ring a parameter element is a non-zerodivisor and cutting by it stays Cohen-Macaulay of dimension one less (algebra chapter, Lemma algebra-lemma-reformulate-CM, canonical tag 02JN). cited at proof lines 2213-2217, 2223-2224 and 2233-2234. t(f) is a non-zerodivisor on the Cohen-Macaulay fibre ring, the cut ring is Cohen-Macaulay, and its dimension is d_2 minus one.
- Grothendieck flatness criterion by fibers (algebra chapter, Lemma algebra-lemma-grothendieck-general, canonical tag 046Z). cited at proof lines 2218-2220. Concludes flatness of the local ring map with Cohen-Macaulay fibre ring, giving part (4).
- Defining diagram of a restriction of a groupoid (more-groupoids chapter, Equation equation-restriction, no canonical tag in the snapshot; no-tag pointer to the source label line). cited at proof line 2226. Identifies the fibre F'_u with the fibre of h over u, reducing part (5) to the computed dimension.

*Source reference.* Original definition 02J1: sources/stacks-chow/tags.txt, source lines 3320–3320. Cited at statement lines 2072-2073: the hypothesis on u is phrased with this notion.

*Source reference.* Original reference situation-slice: sources/stacks-more-groupoids/pinned-original-source.tex, source lines 2041–2041. Cited at statement lines 2075 and 2083: fixes the notation d_1, d_2, h and the conclusion items.

*Source reference.* Original lemma 02J2: sources/stacks-chow/tags.txt, source lines 3321–3321. Cited at proof lines 2095-2097: provides the affine neighbourhood with u closed.

*Source reference.* Original lemma 02FX: sources/stacks-chow/tags.txt, source lines 3208–3208. Cited at proof lines 2128-2129 and 2229-2232: identifies d_2 with the local fibre ring dimension and evaluates the dimension of F'_u at e'(u).

*Source reference.* Original lemma lemma-restrict-preserves-type: sources/stacks-more-groupoids/pinned-original-source.tex, source lines 923–923. Cited at proof lines 2134-2137: the restriction to the point u has locally finite type structure maps.

*Source reference.* Original lemma lemma-groupoid-on-field-dimension-equal-stabilizer: sources/stacks-more-groupoids/pinned-original-source.tex, source lines 1554–1554. Cited at proof lines 2138-2140: dim of the restricted arrow scheme equals dim of its stabilizer.

*Source reference.* Original lemma lemma-groupoid-on-field-locally-finite-type-dimension: sources/stacks-more-groupoids/pinned-original-source.tex, source lines 1515–1515. Cited at proof lines 2141-2142: converts the dimension into the local ring dimension at the identity.

*Source reference.* Original lemma 04ML: sources/stacks-chow/tags.txt, source lines 6047–6047. Cited at proof lines 2143-2145: identifies G" with G_u, giving dimension d_1.

*Source reference.* Original lemma 00NA: sources/stacks-chow/tags.txt, source lines 852–852. Cited at proof lines 2181-2183: each minimal prime quotient of the CM fibre ring has dimension d_2.

*Source reference.* Original lemma 00DS: sources/stacks-chow/tags.txt, source lines 508–508. Cited at proof lines 2192-2194: produces f in the maximal ideal avoiding all minimal primes after t.

*Source reference.* Original lemma 00KW: sources/stacks-chow/tags.txt, source lines 766–766. Cited at proof lines 2195-2197: gives the displayed inequality below d_2.

*Source reference.* Original lemma 02JN: sources/stacks-chow/tags.txt, source lines 3342–3342. Cited at proof lines 2213-2217, 2223-2224 and 2233-2234: non-zerodivisibility of t(f), CM-ness of the cut ring, and the dimension drop by one.

*Source reference.* Original lemma 046Z: sources/stacks-chow/tags.txt, source lines 5490–5490. Cited at proof lines 2218-2220: flatness of the local map with Cohen-Macaulay fibre ring, giving part (4).

*Source reference.* Original equation equation-restriction: sources/stacks-more-groupoids/pinned-original-source.tex, source lines 905–905. Cited at proof line 2226: identifies the fibre F'_u with the fibre of h over u.

**Lemma 0.2.** *Let $S$ be a scheme. Let $(U, R, s, t, c, e, i)$ be a groupoid scheme over $S$. Let $G \to U$ be the stabilizer group scheme. Assume $s$ and $t$ are Cohen-Macaulay and locally of finite presentation. Let $u \in U$ be a finite type point of the scheme $U$, see Morphisms, Definition 02J1. With notation as in Situation situation-slice, set*

$$d_1 = \dim(G_u), \quad d_2 = \dim_{e(u)}(F_u).$$

*If $d_2 > d_1$, then there exist an affine scheme $U'$ and a morphism $g : U' \to U$ such that (with notation as in Situation situation-slice)*

(1) *$g$ is an immersion*

(2) $u \in U'$,
(3) *$g$ is locally of finite presentation,*
(4) *the morphism $h : U' \times_{g,U,t} R \longrightarrow U$ is Cohen-Macaulay at $(u, e(u))$, and*
(5) *we have* $\dim_{e'(u)}(F'_u) = d_2 - 1$.

*Proof.* Let $\mathrm{Spec}(A) \subset U$ be an affine neighbourhood of $u$ such that $u$ corresponds to a closed point of $U$, see Morphisms, Lemma 02J2. Let $\mathrm{Spec}(B) \subset R$ be an affine neighbourhood of $e(u)$ which maps via $j$ into the open $\mathrm{Spec}(A) \times_S \mathrm{Spec}(A) \subset U \times_S U$. Let $\mathfrak{m} \subset A$ be the maximal ideal corresponding to $u$. Let $\mathfrak{q} \subset B$ be the prime ideal corresponding to $e(u)$. Pictures:

$$\begin{array}{ccc} B & \xleftarrow{s} & A \\ {\scriptstyle t}\uparrow & & \\ A & & \end{array} \quad \text{and} \quad \begin{array}{ccc} B_{\mathfrak{q}} & \xleftarrow{s} & A_{\mathfrak{m}} \\ {\scriptstyle t}\uparrow & & \\ A_{\mathfrak{m}} & & \end{array}$$

Note that the two induced maps $s, t : \kappa(\mathfrak{m}) \to \kappa(\mathfrak{q})$ are equal and isomorphisms as $s \circ e = t \circ e = \mathrm{id}_U$. In particular we see that $\mathfrak{q}$ is a maximal ideal as well. The ring maps $s, t : A \to B$ are of finite presentation and flat. By assumption the ring

$$\mathcal{O}_{F_u, e(u)} = B_{\mathfrak{q}}/s(\mathfrak{m})B_{\mathfrak{q}}$$

is Cohen-Macaulay of dimension $d_2$. The equality of dimension holds by Morphisms, Lemma 02FX.

Let $R''$ be the restriction of $R$ to $u = \mathrm{Spec}(\kappa(u))$ via the morphism $\mathrm{Spec}(\kappa(u)) \to U$. As $u \to U$ is locally of finite type, we see that $(\mathrm{Spec}(\kappa(u)), R'', s'', t'', c'')$ is a groupoid scheme with $s'', t''$ locally of finite type, see Lemma lemma-restrict-preserves-type. By Lemma lemma-groupoid-on-field-dimension-equal-stabilizer this implies that $\dim(G'') = \dim(R'')$. We also have $\dim(R'') = \dim_{e''}(R'') = \dim(\mathcal{O}_{R'', e''})$, see Lemma lemma-groupoid-on-field-locally-finite-type-dimension. By Groupoids, Lemma 04ML we have $G'' = G_u$. Hence we conclude that $\dim(\mathcal{O}_{R'', e''}) = d_1$.

As a scheme $R''$ is

$$R'' = R \times_{(U \times_S U)} \Big( \mathrm{Spec}(\kappa(\mathfrak{m})) \times_S \mathrm{Spec}(\kappa(\mathfrak{m})) \Big)$$

Hence an affine open neighbourhood of $e''$ is the spectrum of the ring

$$B \otimes_{(A \otimes A)} (\kappa(\mathfrak{m}) \otimes \kappa(\mathfrak{m})) = B/s(\mathfrak{m})B + t(\mathfrak{m})B$$

We conclude that

$$\mathcal{O}_{R'', e''} = B_{\mathfrak{q}}/s(\mathfrak{m})B_{\mathfrak{q}} + t(\mathfrak{m})B_{\mathfrak{q}}$$

and so now we know that this ring has dimension $d_1$.

We claim this implies we can find an element $f \in \mathfrak{m}$ such that

$$\dim(B_{\mathfrak{q}}/(s(\mathfrak{m})B_{\mathfrak{q}} + fB_{\mathfrak{q}}) < d_2$$

Namely, suppose $\mathfrak{n}_j \supset s(\mathfrak{m})B_{\mathfrak{q}}$, $j = 1, \ldots, m$ correspond to the minimal primes of the local ring $B_{\mathfrak{q}}/s(\mathfrak{m})B_{\mathfrak{q}}$. There are finitely many as this ring is Noetherian (since it is essentially of finite type over a field – but also because a Cohen-Macaulay ring is Noetherian). By the Cohen-Macaulay condition we have $\dim(B_{\mathfrak{q}}/\mathfrak{n}_j) = d_2$, for example by Algebra, Lemma 00NA. Note that $\dim(B_{\mathfrak{q}}/(\mathfrak{n}_j + t(\mathfrak{m})B_{\mathfrak{q}})) \leq d_1$ as it is a quotient of the ring $\mathcal{O}_{R'', e''} = B_{\mathfrak{q}}/s(\mathfrak{m})B_{\mathfrak{q}} + t(\mathfrak{m})B_{\mathfrak{q}}$ which has dimension $d_1$. As $d_1 < d_2$ this implies that $\mathfrak{m} \not\subset t^{-1}(\mathfrak{n}_i)$. By prime avoidance, see Algebra, Lemma

00DS, we can find $f \in \mathfrak{m}$ with $t(f) \notin \mathfrak{n}_j$ for $j = 1, \ldots, m$. For this choice of $f$ we have the displayed inequality above, see Algebra, Lemma 00KW.

Set $A' = A/fA$ and $U' = \mathrm{Spec}(A')$. Then it is clear that $U' \to U$ is an immersion, locally of finite presentation and that $u \in U'$. Thus (1), (2) and (3) of the lemma hold. The morphism

$$U' \times_{g,U,t} R \longrightarrow U$$

factors through $\mathrm{Spec}(A)$ and corresponds to the ring map

$$B/t(f)B \xleftarrow{=\!=} A/(f) \otimes_{A,t} B \xleftarrow{s} A$$

Now, we see $t(f)$ is not a zerodivisor on $B_{\mathfrak{q}}/s(\mathfrak{m})B_{\mathfrak{q}}$ as this is a Cohen-Macaulay ring of positive dimension and $f$ is not contained in any minimal prime, see for example Algebra, Lemma 02JN. Hence by Algebra, Lemma 046Z we conclude that $s : A_{\mathfrak{m}} \to B_{\mathfrak{q}}/t(f)B_{\mathfrak{q}}$ is flat with fibre ring $B_{\mathfrak{q}}/(s(\mathfrak{m})B_{\mathfrak{q}} + t(f)B_{\mathfrak{q}})$ which is Cohen-Macaulay by Algebra, Lemma 02JN again. This implies part (4) of the lemma. To see part (5) note that by Diagram (equation-restriction) the fibre $F'_u$ is equal to the fibre of $h$ over $u$. Hence $\dim_{e'(u)}(F'_u) = \dim(B_{\mathfrak{q}}/(s(\mathfrak{m})B_{\mathfrak{q}} + t(f)B_{\mathfrak{q}}))$ by Morphisms, Lemma 02FX and the dimension of this ring is $d_2 - 1$ by Algebra, Lemma 02JN once more. This proves the final assertion of the lemma and we win. □